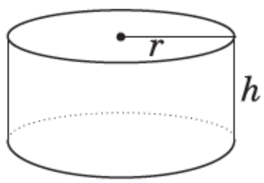
Part 4: Maximize the Volume of a Cylinder

Cylinder 	$A_{\text{base}} = \pi r^2$ $A_{\text{lateral surface}} = 2\pi r h$ $A_{\text{total}} = 2A_{\text{base}} + A_{\text{lateral surface}}$ $= 2\pi r^2 + 2\pi r h$	$V = (A_{\text{base}})(\text{height})$ $V = \pi r^2 h$

Investigation 1: Maximize Volume of a Cylinder for a Given Surface Area

Your task is to design a cylindrical juice can that uses 470cm^2 of aluminum. The can should have the greatest volume possible.

a) Substitute the given surface area and radius into the surface area formula and solve for the height, h. You can use the rearranged formula $h = \frac{SA - 2\pi r^2}{2\pi r}$

b) Determine the volume of this can using the formula for the volume of a cylinder: $V = \pi r^2 h$. Record the data in the following table:

Surface Area	Radius	Height	Volume
470	2		
470	5		
470	8		

c) What is the maximum volume for the cans in your table? What are the radius and height for this can?

Conclusion: The maximum volume for a given surface area of a cylinder occurs when its height equals its diameter. That is, $h =$ ___ or $h = 2$___

Example 1:

a) Determine the dimensions of the cylinder with maximum volume that can be made with 600cm^2 of aluminum. Round the dimensions to the nearest hundredth of a centimeter.

Solution:

We know that a cylinder with height equal to the diameter (double the radius) gives the maximum volume for a given surface area.

To determine the radius of the can with a given surface area, use the surface area formula and substitute h = 2r.

Given: $SA = 2\pi r^2 + 2\pi rh$ and $h = diameter = 2r$

$$SA = 2\pi r^2 + 2\pi r(2r)$$
$$SA = 2\pi r^2 + 4\pi r^2$$
$$SA = 6\pi r^2$$

To solve for the radius given a surface area, you must use the rearranged formula: $r = \sqrt{\frac{SA}{6\pi}}$

$$r = \sqrt{\frac{600}{6\pi}} = 5.64$$

The radius of the cylinder should be 5.64 cm and the height should be twice that, or 11.28 cm.

b) Determine the volume of this cylinder

$$Volume = \pi r^2 h$$
$$Volume = \pi(5.64)^2(11.28)$$
$$Volume = 1127\ cm^3$$

Apply What You Learned

1. Determine the dimensions of the cylinder with the maximum volume for each surface area. Round the dimensions to the nearest hundredth of a unit.

a) 1200 cm^2

b) 125 cm^2

2. Determine the volume of each cylinder in question 1. Round to the nearest whole unit.

a)

b)